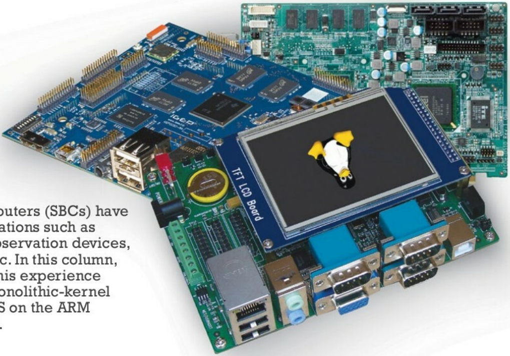
Develop a GNU/Linux-like OS for a Single Board Computer



Single board computers (SBCs) have a variety of applications such as monitoring and observation devices, kiosk terminals, etc. In this column, the author shares his experience of developing a monolithic-kernel GNU/Linux-like OS on the ARM platform for a SBC.

Single board computers (SBCs) have become pretty popular in a wide variety of fields. As the core component of computer systems as well as of embedded systems, the operating system plays a very important role in these systems.

For the purpose of technical research and teaching a curriculum, I have developed a monolithic-kernel SBC GNU/Linux-like OS on the ARM platform. The article covers a boot loader design called U-boot, building the kernel – uImage, the design of the root file system and the Init process. The single board computer OS (SBC OS) is developed on the Linux platform with the GNU tool chain. The system mainly focuses on helping students to learn about and design tiny operating systems on the ARM platform from scratch when the source code is provided.

Architecture of the SBC OS

At the top of the SBC OS is the user or application space where user applications are executed. Below the user space is the kernel space, where the SBC OS kernel resides.

The SBC OS also contains a GNU C library *(glibc)*, which provides the system call interface that connects to the SBC OS kernel and provides the mechanism for the transition between the user or application space and the SBC OS kernel. This is important because the kernel and the user application occupy different protected address spaces. While each user or application space process occupies its own virtual address space, the SBC OS kernel occupies a single address space.

The SBC OS kernel can be further divided into three levels. At the top is the system call interface, which implements the basic functions such as read and write. Below the system call interface is the SBC OS kernel code, which can be more accurately defined as the architecture-independent kernel code. This code is common to all of the processor architectures supported by the SBC OS. Below this is the architecture-dependent code, which forms what is more commonly called a BSP (board support package). This code serves as the processor and platform-specific code for the given architecture.

Design and implementation

The U-boot boot loader design

U-boot is an open source, cross-platform boot loader that provides out-of-the-box support for hundreds of SBCs and many CPUs, including PowerPC, ARM, XScale, MIPS, Coldfire, NIOS, Microblaze and x86.

The SBC OS normally resides in large-capacity devices such as hard disks, CD-ROMs, USB disks, network servers and other permanent storage media. When the processor is powered on, the memory does